Francisco Samagaio

Junior Data Scientist
Porto, Portugal

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LinkedIn | GitHub | Portfolio

Machine Learning | Python | SQL | Generative AI | Data Analytics

Work Experience

Junior Data Analyst — Natixis in Portugal
Dec 2024 – Present

- Developed and optimized SQL queries across ~300 financial reports in a large-scale Oracle data lake, improving performance and ensuring data accuracy.
- Designed, maintained, and validated 30+ database tables supporting ETL pipelines and reporting workflows.
- Built automated reporting and data transformation solutions using EasyMorph for stakeholder-driven analytics.
- Led migration of ~80 reports from IBM Cognos to EasyMorph, rewriting and optimizing SQL logic for improved performance and maintainability.
- Collaborated in a regulated financial environment (CIB), strengthening experience in large-scale financial data processing, reporting, and analytics delivery.
- Gained domain knowledge in Corporate & Investment Banking products, reporting structures, and regulatory-driven data workflows.

Intern Developer — Natixis in Portugal
Dec 2023 – Dec 2024

- Developed Python-based automation pipelines replacing legacy PowerShell workflows, contributing to modernization of internal systems.
- Built ETL pipelines using Pandas and NumPy, processing 150+ multi-format files (CSV, Excel, TXT, JSON).
- Automated multi-source data integration workflows, reducing manual processing effort and improving efficiency.
- Worked with Git, Jenkins, and Linux in CI/CD environments supporting production systems.
- Contributed to batch processing and financial data systems using COBOL, JCL, and CICS.

Research Intern / Master's Thesis — Fraunhofer Portugal AICOS
Feb 2023 – Jul 2023

- Developed a Master's thesis on generative AI for automated ceramic color generation in product design.
- Fine-tuned deep learning models using PyTorch and explored state-of-the-art image generation architectures.
- Worked with Python, HPC environments, Linux, and Jupyter for large-scale experimentation.
- Evaluated generative models using visual quality metrics and user studies, contributing to applied AI research in industrial design.

Technical Skills

Programming: Python, SQL, C/C++
Machine Learning & AI: PyTorch, Scikit-learn, Generative AI, Model Evaluation, Feature Engineering
Data Analysis: Pandas, NumPy, Data Cleaning, Statistical Analysis
Data Engineering: ETL Pipelines, Data Lakes, SQL Optimization
Tools: Git, Linux, Jupyter, Jenkins, Bash

Education

Faculty of Engineering of the University of Porto — Master's Degree in Electrical and Computer Engineering Master's Thesis: Generative AI for Ceramic Color Generation	**2018 - 2023**
University of Genoa — Erasmus+ Study Program	**2022**

Languages

Portuguese — Native; **English** — Fluent; **Spanish** — Intermediate; **French** — Beginner